HepatoTwin

Clinical Safety, Explanations, In-Silico Evaluation and Thesis Writing

Progress Report — Meeting 10
Date: 30th October 2026

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1. Project Objective

Meeting 10 covered the two tasks of the weekly action plan, the first part of the in-silico evaluation, and thesis writing:

1. **Task 1, twin state and evidence:** fibrosis risk and a referral pathway, disease subtype and safety guardrails, per-patient explanations and uncertainty, a causal knowledge graph, NHANES III data for external validation, and the virtual-patient simulator (in-silico evaluation, part 1).
2. **Task 2, recipe ranking and explanations:** recipe explanations traced through the knowledge graph, grounded plain-language summaries with an automatic faithfulness check, and improved cuisine and feasibility scores.
3. **Thesis writing:** the Introduction and Literature Review chapters.

All evaluations use held-out patients aged 12–59, the scope of the Layer 2 model; fibrosis scoring and the NHANES III data cover adults only.

2. Work Completed

2.1 Task 1: Twin State and Evidence

- Implemented FIB-4 fibrosis risk with the EASL-EASD-EASO 2024 cut-offs and validated it against liver stiffness measured in the same FibroScan examination; a high score withholds recipes until a clinical assessment.
- Implemented MASLD, MetALD and alcohol-related liver disease labels from the 2023 nomenclature, and safety notices for patients under 18, with higher alcohol intake, taking glucose-lowering medicine or pregnant.
- Added per-patient SHAP explanations of the predicted CAP, a 90% conformal CAP interval and bootstrap bands for the calibrated probabilities.
- Built a causal knowledge graph of 14 links that restricts dietary priorities to causes of liver fat.
- Downloaded the NHANES III examination, laboratory and hepatic ultrasound files and mapped them onto the model's inputs.
- Built the virtual-patient simulator with cited evidence tables (in-silico evaluation, part 1).

2.2 Task 2: Recipe Ranking and Explanations

- Explained each recommended recipe through the knowledge graph, from each criterion to liver fat with its evidence rows, plus the cuisine and practicality context.
- Generated patient summaries from a template using only the twin's own facts, with an automatic check of every number and citation and a slot for a language model later.

- Added ingredient co-occurrence to the cuisine score and cooking techniques and equipment to the feasibility score, and re-evaluated the ranking on 300 held-out patients.

2.3 Thesis Writing

- Drafted the Introduction (Chapter 1) and Literature Review (Chapter 2) in LaTeX, with preview PDFs.
- Covered every paper in the "papers read" tracker; about 60 papers were read in full and the rest from their abstracts.
- Checked published R^2 values against the full papers and corrected six figures in earlier reports.

Implementation status: All code is maintained in the project repository, and 129 automated tests pass (84 from Meeting 9 and 45 new).

3. Technical Implementation and Results

3.1 Fibrosis Risk and Referral Pathway

FIB-4 = age × AST / (platelets × √ALT) (Sterling et al., 2006) is computed for patients aged 18–59: below 1.3 is low risk, 1.3–2.67 indeterminate (a second test such as elastography is advised) and above 2.67 high risk. Patients under 18 receive no score, because FIB-4 was developed in adults. It was validated against liver stiffness in 7,586 adults aged 18–59 with a reliable reading (IQR/median ≤ 30%).

Table 1. FIB-4 against measured liver stiffness, adults aged 18–59.

Stiffness	Prevalence	AUC	Sensitivity (FIB-4 ≥ 1.3)	NPV (FIB-4 < 1.3)	Specificity (> 2.67)	PPV (> 2.67)
≥ 8 kPa	8.9%	0.593	0.199	0.922	0.996	0.537
≥ 12 kPa	3.0%	0.668	0.330	0.978	0.994	0.388

90.8% of adults were low risk, 8.3% indeterminate and 0.9% high risk. FIB-4 rules out raised stiffness well (negative predictive value 0.978 for ≥ 12 kPa) but detects it poorly (AUC 0.593 and 0.668), which matches its known weakness in young, general populations. It is therefore used as a rule-out test.

Two refinements follow from this. Because FIB-4 is less reliable under age 35 (AASLD 2023 Practice Guidance), a low score at ages 18–34 now carries a note that it does not fully rule out scarring and that a FibroScan would help; this applies to 35.6% of adults. And when a patient has a reliable FibroScan reading, for example after referral, the measured stiffness replaces FIB-4: below 8 kPa low risk, 8–12 kPa indeterminate, and 12 kPa or more high risk, with recipes withheld until a clinical assessment. NHANES participants are evaluated without their stiffness, as a real user without a scan would be.

3.2 Disease Subtype and Safety Guardrails

Steatosis with at least one of five cardiometabolic criteria (body size, glycaemia, blood pressure, triglycerides, HDL) is MASLD; MASLD with 140–350 g alcohol a week (women) or 210–420 g (men) is MetALD; above that range the label is alcohol-related liver disease (ALD). Alcohol is estimated from the NHANES frequency and quantity questions at 14 g per US standard drink.

Table 2. Labels in the in-scope cohort, using measured CAP ≥ 248 dB/m as steatosis.

Label	Adults (18–59)	Adolescents (12–17)
MASLD	51.1%	24.5%
MetALD	3.1%	0%
ALD	1.3%	0%
Steatosis without a cardiometabolic factor	0.6%	2.4%
No steatosis	43.8%	73.1%

Safety notices: 17.4% of in-scope participants are under 18, 5.8% drink in the MetALD range or above, and 6.2% take glucose-lowering medicine. No pregnant participant is in the cohort, because NHANES does not perform FibroScan in pregnancy; the notice remains for real users. A high FIB-4 adds a referral notice and withholds recipes.

3.3 Patient-Level Explanations and Uncertainty

Per-patient SHAP values split each predicted CAP into the average prediction plus one contribution per input, in dB/m. For participant SEQN 109349 (predicted CAP 313.7 dB/m against an average of 258.0), waist circumference added 15.6 dB/m and the derived Fatty Liver Index 10.5 dB/m.

Table 3. Conformal CAP intervals: held-out test coverage within the scope.

Target coverage	Interval	Observed coverage	Ages 12–17	Ages 18–39	Ages 40–59
80.0%	± 50.5 dB/m	80.1%	83.3%	80.5%	78.4%
90.0%	± 66.6 dB/m	90.9%	93.8%	90.8%	89.7%
95.0%	± 82.3 dB/m	95.6%	96.7%	95.1%	95.5%

The 90% interval is wide (± 66.6 dB/m), the direct consequence of the model's R^2, which is why the twin leads with calibrated probabilities. Bootstrap bands for those probabilities have a mean width of 5.4 percentage points (CAP ≥ 248 dB/m) and 5.1 (CAP ≥ 280 dB/m); they cover the calibration step only.

3.4 Causal Knowledge Graph

Fourteen hand-written links connect six levers (energy, saturated fat, added sugar, diet quality, physical activity, alcohol) through body weight and insulin resistance to liver fat, liver injury and fibrosis; CAP, ALT and liver stiffness are measurements only. Each link cites rows of the evidence table or the paper it rests on. A dietary priority may act only on a lever with a causal path to liver fat, so no recipe criterion targets a measurement such as ALT or CAP.

Table 4. Causal chains behind the four dietary priorities.

Priority	Chain to liver fat	Evidence (strongest link)
Weight management	energy intake → body weight → liver fat	VG2015-WL5 (STRONG)
Limit saturated fat	saturated fat intake → liver fat	LUU2018-SAT (MODERATE)
Limit excess sugar	added sugar intake → liver fat	CHI2014-HYPER (MODERATE)
Diet quality	Mediterranean-style diet quality → liver fat	STE2026-MEDDIET (MODERATE)

3.5 NHANES III External Validation Data

The NHANES III (1988–1994) examination and laboratory files and the 2009–2010 re-reading of the archived gallbladder ultrasound for hepatic steatosis were downloaded and mapped onto the model's inputs, converting units where they differ (for example calcium from mmol/L and CRP from mg/dL).

- 10,682 adults aged 20–59 with a gradable image; 20.7% have moderate or severe steatosis.
- 51 of the 81 model inputs are mapped; the 24-hour recall, questionnaire items and some blood counts are not, and the model treats them as missing.
- Mapped values agree with the modern cohort except where the population has changed (median BMI 26.1 against 28.4 kg/m²); assays were not recalibrated, and NHANES III CRP is not a high-sensitivity assay.
- The public-use Linked Mortality File links 10,672 of them to deaths up to 31 December 2019: 2,174 deaths over a median 27.3 years. Its cause-of-death groups have no liver category, so it supports all-cause, cardiovascular and cancer mortality but not liver-related deaths.

3.6 Cuisine and Feasibility Scores

In RecipeDB, 7,134 of the 10,108 candidate meals (71%) are labelled "Other" and previously scored 0 for every cuisine. Ingredient co-occurrence now measures how typical a recipe's ingredient pairs are of a cuisine: recipes labelled with a cuisine score a median of 55–66 for it, against 11–20 for recipes labelled with other cuisines. Feasibility now also reads the instructions: each demanding technique (deep-frying, kneading dough, overnight marinating, thermometer work, flambéing) lowers the score by 25% and each piece of special equipment by 10%.

Table 5. Default ranking on 300 held-out patients: share of top-10 recipes meeting each criterion.

Cuisine scoring	Health	Cuisine label	Feasible	All three	Ingredient typicality	Demanding technique
Meeting 9: label only; time, ingredients, steps	1.000	0.998	0.832	0.831	–	–
Label and ingredients blended; techniques averaged in	1.000	0.865	0.850	0.742	75	0.0%
Ingredients fill label gaps; techniques averaged in	1.000	0.980	0.793	0.793	55	0.0%
Adopted: ingredients fill label gaps; techniques as a penalty	1.000	0.980	0.822	0.822	53	0.0%

Four designs were compared. Blending the label with the ingredients for every recipe narrowed the gap between a correctly labelled recipe and an unlabelled one from 100 to about 10 points, so practicality outweighed cuisine and fewer top-10 recipes were in the patient's cuisine. Scoring only unlabelled recipes on ingredients (up to 50) restored the cuisine match. Averaging the technique check in as a fourth feasibility component diluted time, ingredients and steps and lowered the practical share; applying it as a penalty leaves recipes without demanding techniques unchanged. The adopted design combines the two.

3.7 Recipe Explanations and Plain-Language Summaries

Each of the top three recipes is explained criterion by criterion: the value, the reference it meets, the causal chain to liver fat with evidence rows and strength, and the cuisine and practicality context. For

SEQN 109349 the first recipe, Chipotle Pinto Beans, has 0.8% of energy from saturated fat against a reference of 10%, traced as "saturated fat intake → liver fat (MODERATE, LUU2018-SAT, EASL2024-DIET)".

The patient summary is written by a template from the twin's own facts: the CAP estimate and interval, the probabilities, the label, FIB-4, safety notices, priorities with evidence identifiers, and recipe names. An automatic check confirms that every number and citation in the text appears in those facts. A language model can be plugged in later; if its text fails the check, the template is shown instead. Tests confirm that an invented number or citation is caught and that recipes are left out when they are withheld.

3.8 In-Silico Evaluation Design

The evaluation has two separate parts. The simulator plays the patient: it generates each virtual patient's true monthly weight, blood tests, liver fat and CAP from published trials, and the measurements a real patient would report. The twin (Layers 2–5) sees only those measurements and must estimate, predict and adjust each month. Comparing what the twin estimated with what the simulator knew shows how well it tracks, detects change, projects and learns each patient's response.

- **Independence:** the simulator never uses the Layer 2 model, so the twin is not graded against its own output. An automated test enforces this.
- **Purpose:** synthetic data are used to test the twin, never to train it.
- **Terminology:** the design is an in-silico evaluation with virtual patients (a plasmode simulation, or a "twin experiment" in data assimilation).
- **What it can show:** that the twin updates, tracks and adapts correctly when the truth is known; it does not show that the twin predicts real patients.

3.9 Evidence for the Simulator

Every number the simulator uses is stored in a cited table, with its source, the location in the paper, and whether it was quoted or derived (with the calculation written out).

Table 6. Simulator parameter table by level of access.

Access	Rows	Meaning
Full text	33	Read in the full paper
Abstract	15	Only the abstract was available
Database	9	EFLM Biological Variation Database; CDC growth charts
Design	6	Project assumptions, stated as such
Secondary	1	From a review; primary trial still to be confirmed

Table 7. Calibration targets the virtual patients must reproduce.

Trial	Targets	Use
Look AHEAD, year 1	Mean weight change −8.6% (intensive) vs −0.6%; ≥ 5% lost: 68.0% vs 13.3%	Tune
Look AHEAD, year 8	Mean −4.7%; ≥ 5% lost: 50.3%; ≥ 10% lost: 26.9%	Held back
Diabetes Prevention Program	50% reached ≥ 7% loss at 24 weeks; 38% kept it	Tune; held back
Vilar-Gomez et al. (2015)	30% lost ≥ 5% at 52 weeks with ordinary lifestyle advice	Tune

Trial	Targets	Use
Sangprasert et al. (2025)	CAP 319 → 292 dB/m over 6 months	Held back

Held-back targets are not looked at until calibration is complete, so they remain an honest check.

3.10 Virtual-Patient Simulator

Month 0 is the 1,236 real in-scope test patients. Each follows the usual diet, scenario A (−500 kcal/day) or scenario B (−250 kcal/day) for 12 months. A person's adherence pattern (steady, slipping, regaining or dropping out) and how strongly they respond are drawn once per person, so scenarios for the same person differ only in diet.

- **Weight:** Hall's energy-balance rule (1 kg per 100 kJ/day eventually), with each patient's speed and eventual loss set by their body fat (time constant 5–24 months).
- **Growth:** patients aged 12–17 (209) grow along their own CDC weight and height centiles.
- **Blood markers:** glucose, HbA1c, insulin, lipids and blood pressure change by a share of the patient's own baseline, with separate evidence for patients with diabetes (Look AHEAD) and without (34 pooled trials); CRP, uric acid and white cells change per kg.
- **Liver:** liver fat falls with weight lost and further while eating less; CAP follows liver fat (5 dB/m per PDFF point); ALT, AST and GGT follow the liver-fat change.

Table 8. True trajectories of the virtual cohort (calibration not yet applied).

Scenario	Month	Mean weight change	Lost ≥ 5%	Lost ≥ 10%	Mean CAP change (dB/m)
A (−500 kcal/day)	6	−5.1%	44%	6%	−9.5
A (−500 kcal/day)	12	−5.9%	50%	17%	−11.3
B (−250 kcal/day)	6	−2.3%	5%	0%	−4.8
B (−250 kcal/day)	12	−2.5%	17%	1%	−5.8
Usual diet	12	+1.0%*	0%	0%	0.0

**Growth of the adolescents; adults on the usual diet do not change.*

Table 9. Weaknesses found in the first run and how they were fixed.

Problem	Fix	Evidence
CAP conversion five times too large	Use the published CAP cut-offs (275 dB/m ≈ 10% PDFF): 5 dB/m per point	Wang et al. (2020)
ALT fell too much in people with healthy livers	Liver enzymes follow the change in liver fat	Koutoukidis et al. (2019); Wong et al. (2013)
Blood-sugar effects came from people with diabetes	Separate evidence for patients without diabetes	Appuhamy et al. (2014)
Liver fat reacted too slowly	Extra fall while eating less, fitted to the first week of Lim et al.	Lim et al. (2011); Patel et al. (2015)
One weight-loss speed for everyone	Personal speed and eventual loss from body fat	Hall (2011) web appendix; Deurenberg et al. (1991)
Adolescents did not grow	Growth along CDC centiles	CDC 2000 growth charts
Insulin, hip, arm, CRP fixed	Each now changes with weight	Appuhamy (2014); Lim (2011); Selvin et al. (2007)
Uric acid and white cells fixed	Each now changes per kg lost	Yokose et al. (2020); Dixon & O'Brien (2006)

A workaround for markers without trial evidence was tested and rejected: estimating a weight-loss effect from how the marker differs between heavier and lighter people in NHANES. For blood tests with known trial values it was off by a factor of 0.6 to 5.5, and in the wrong direction for HDL. It is used only for arm circumference, where the same method reproduced the trial values for hip (0.63 vs 0.65 cm/kg) and waist (0.77 vs 0.86 cm/kg). Red-cell measures, platelets and ferritin are held constant: no trial shows a change, and they barely differ with body size.

3.11 Thesis Chapters

The Introduction (about 1,750 words) covers the background, digital twins, computational gastronomy, the problem and research gap, the research question and objectives, the approach with a seven-layer overview figure, contributions, scope and limitations, and the thesis outline.

The Literature Review (about 4,900 words, 84 references) covers MASLD and its dietary treatment, non-invasive measurement of liver fat, machine-learning prediction and its limits, dietary assessment, digital twins, computational gastronomy and food recommendation, and how a twin can be evaluated without longitudinal data. It ends with a table of the research gap.

The full-text check corrected six figures in the R^2 explanation document, Experiment Record and Progress Report: the published range (0.34–0.56 → 0.34–0.54), FLI+ (0.56 for one fold → 0.54, the four-fold mean), FLI alone (0.28–0.35 → 0.28–0.36), UK Biobank DXA (0.63–0.65 → 0.63–0.64), Haas ("not reliable" → r 0.58) and Zhu et al. (AUC 0.892, a precision–recall value → macro AUC 0.874 across three grades).

About 60 papers were read in full. About 22 could only be read as abstracts because they are behind paywalls; these are cited only for what their abstracts state. Published R^2 values appear in the Literature Review; the project's own R^2 experiments will go in the planned Experiments chapter.

4. Current Challenges and Limitations

- **FIB-4 detects fibrosis poorly in this population:** it is used to rule out raised stiffness, not to find it.
- **Adolescent MASLD criteria:** checked against the nomenclature's figure. Two alternatives are not used: waist above the 95th percentile and, at age 12, blood pressure above the 95th percentile, because they need reference tables the project does not have. They can only miss a criterion, never add one.
- **NHANES limits for the labels:** blood-pressure and cholesterol treatment are not available, so a diagnosis stands in for treatment; triglycerides are non-fasting; other causes of steatosis are not assessed.
- **Wide CAP intervals:** the 90% interval of about ± 67 dB/m reflects the model's accuracy; probabilities remain the twin's main state.
- **Cuisine labels:** only 29% of candidate meals carry a specific cuisine label; ingredient co-occurrence fills gaps but is an indirect signal.
- **NHANES III:** assays and the population differ from 2017–2023, the reference is an ultrasound grade rather than CAP, and the public mortality file cannot identify liver-related deaths.
- **Summaries:** the template is safe but plain and lists evidence identifiers; a language model has not yet been connected.
- **Simulator not yet calibrated:** adherence patterns and response spread are starting guesses until they are tuned to the trial targets, and scenarios A and B are fixed calorie reductions.

- **User-retrieved values:** the EFLM biological-variation values and the DIRECT uric-acid coefficient were read from the source by the user; screenshots should be kept for the appendix.

5. Decisions Taken During Meeting 10

5.1 Use FIB-4 as a rule-out test with a referral before recipes

FIB-4 rules out raised stiffness reliably but detects it poorly. High-risk patients are asked to seek a clinical assessment, and recipes are withheld until then.

A measured FibroScan stiffness, when available, replaces FIB-4, and a low FIB-4 under age 35 carries a note recommending one.

5.2 Restrict dietary priorities to causes of liver fat

The causal knowledge graph allows a recipe criterion only on a lever with a causal path to liver fat, so the twin never optimises a measurement such as ALT or CAP directly.

5.3 Let ingredients fill cuisine gaps without overriding labels

Blending label and ingredients for every recipe lowered the share of top-10 recipes in the patient's cuisine. Labelled recipes therefore keep their label score, and only unlabelled recipes are scored on ingredients.

5.4 Use template summaries until a language model is chosen

Summaries are written from a template now; a language model can be added later through the same automatic faithfulness check, with the template as the fallback.

5.5 Keep the simulator independent and use it only for testing

The simulator is built only from published evidence and never uses the Layer 2 model. Synthetic data test the twin and are never used to train it.

5.6 Fix weaknesses with evidence rather than list them as limitations

Eight weaknesses in the simulator's first run were fixed using new published sources. A shortcut for markers without evidence was tested first and rejected because it failed on markers with known answers.

6. Overall Project Status

All tasks planned for Meeting 10 are complete. The main outcomes are:

- A second dimension of the twin state: fibrosis risk, validated against measured stiffness, with a referral pathway.
- Research labels for MASLD, MetALD and ALD, and safety notices for groups for whom unsupervised diet change may not be appropriate.
- Per-patient explanations, a 90% CAP interval with verified coverage, and probability bands.
- A causal knowledge graph that both constrains the priorities and explains each recipe.
- Template summaries with an automatic faithfulness check and a slot for a language model.
- Cuisine and feasibility scores that use ingredient co-occurrence and cooking techniques.
- NHANES III data mapped for external validation and linked to 27 years of mortality follow-up.
- A working virtual-patient simulator and the first two thesis chapters.
- 129 passing automated tests.

7. Immediate Next Steps — Meeting 11

From the weekly action plan:

1. **Backend:** a FastAPI service wrapping Layers 2–5 with an SQLite database.
2. **Re-assessment timeline:** dated measurements and food logs with the predict–observe–compare–adjust loop.
3. **3D liver view:** grades S0–S3 with a timeline slider, coloured by the calibrated probabilities and their uncertainty.
4. **In-silico evaluation, part 2:** the observation layer (laboratory variation, CAP noise, under-reported food logs, quarterly blood tests, missed visits) and calibration of the simulator to the trial targets.
5. **Dashboard and new criteria:** the React dashboard, ultra-processing (FoodProX) and glycaemic load.

Also outstanding: continue recruiting 5–10 pilot evaluators.